Tower-Coordinate-Free Road-Level Trajectory Recovery from Cell-ID Sequences: Supplementary Material

Xiaoxiao Wu*, Xiao Zheng*, Hairong Zhou*, Yongjie Zhu†
* College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China
Emails: xxwu.eesissi@szu.edu.cn, m15649741159@163.com, zhr19466@gmail.com
† Nexwise Intelligence China Limited, Guangzhou, China
Email: zhuyj@nexwise.com.cn

Reviewer Navigation

This concise supplement contains the material most useful for checking implementation and empirical claims: **A**, the information boundary, evidence artifact, graph contract, and metrics; **B–E**, the exact PPO state, reward aggregation, optimization, and training artifact audits complementing the main-paper training contract; **F**, online time-aware SVAA/PPO-guided MCTS; **G**, controlled baseline and ablation definitions plus aggregate results; **H**, method and qualitative figures; and **I**, the cross-city pipeline-transfer and controlled million-record scaling lab (Region B) alongside supplementary audit protocols. The oracle construction, reward/boundary table, bounded training lookahead, and soft corridor equations now appear in main-paper Section III-D; data scaling and multi-seed inference repeatability appear in main-paper Section IV-E and Fig. 4. They are not reproduced here. **J** gives the reviewer checklist and limitations. The full 13-sheet route gallery and per-trajectory tables are excluded from this reviewer-facing PDF; the all-trajectory ablation audit is retained as a fixed-order heatmap.

Terminology. We call the proposed sequence module *Set-Valued Asynchronous Alignment (SVAA)*. SVAA computes a maximum-cardinality ordered matching from Cell-ID observation positions to nonempty, nonoverlapping, contiguous path-edge intervals. A set-valued LCS recurrence is only its deterministic anchor step; SVAA then expands compatible intervals without overlap. Unlike ordinary LCS, compatibility is many-to-many rather than token equality; unlike synchronous matching, SVAA permits skipped observations/edges and lets one observation explain a multi-edge interval. Throughout this supplement, *SVAA witness* means this deterministic output. The main paper's "CID/LCS progress" features are precisely its cardinality and latest increment.

A. Inputs, Evidence, and Deterministic Graph Contract

Information boundary.: Regional calibration uses the probe's GPS–CID records, session/trace identifiers, original acquisition order, and a frozen directed road graph. The supplied coordinates have already been de-duplicated. Evidence-cache construction, training-data generation, the training-only coordinate oracle, PPO training, and bounded training search use no timestamps. Online prediction receives the frozen graph, endpoints, a timestamped raw-CID sequence, evidence cache, PPO checkpoint, and matching normalizer. Query timestamps determine the online hard temporal constraints and soft pair-time/corridor terms used by action priors, leaf evaluation, and final ranking. Intermediate query GPS and reference edges are withheld from prediction. Held-out test records are reserved for final evaluation.

Raw-CID token and sequence construction.: The model token is the normalized raw CID alone. Public land mobile network (PLMN) and radio access technology (RAT) are retained only for acquisition audit, not as model inputs or cache keys. In Region A, 295 of 911 raw-CID values occur under multiple PLMN namespaces, all using Long-Term Evolution (LTE); the system intentionally pools records sharing a raw CID into one empirical support Ω_c. It neither asserts one physical cell nor performs operator identification or cross-operator cell disambiguation. Each retained record is the `Index`=1 cell from one logical scan channel—the channel's first-ranked scan result, not an asserted serving or camped cell. Records form one sequence. Equal-time records retain their deterministic device-output order and equal timestamp, without implying physical switch chronology. All CID-consuming methods receive the same permitted raw query information. Yen, Beam, and the proposed decoder share the query-sequence and evidence contract above; other baselines may apply their documented internal pooling or subsampling. Common raw information therefore does not require identical internal representations. No rank weighting or multi-stream fusion is used by the proposed decoder.

A.1 Road–Cell-ID Evidence Artifact

Accepted calibration rows contain CID, longitude, latitude, and either a trace identifier or a single-trace-per-file declaration. Coordinates have already been de-duplicated in the supplied base data; repeated CID labels at different retained positions remain distinct records. Malformed or nonfinite required fields are errors. Processing uses fixed trace and original within-trace row order for reproducible reductions and geometric

displacements. No timestamp is required or used for sorting, grouping, coordinate de-duplication, filtering, or evidence aggregation.

Let (λ_0, ϕ_0) be the mean accepted longitude and latitude in degrees. Map every point and road vertex to metres by

$$x = (\lambda - \lambda_0)\cos(\pi\phi_0/180)\,111320,$$
$$y = (\phi - \phi_0)\,111320.$$

Thus the trigonometric argument is in radians while longitude and latitude differences remain in degrees. For CID c, bin points into 80-metre cells, retain cells in decreasing count (then increasing cell index) through at least 90% cumulative mass, and buffer their union by 50 metres to form the closed support Ω_c. With population variances over all N_c records,

$$\sigma_c = \min\{800, \max\{80, \sqrt{\mathrm{Var}(x) + \mathrm{Var}(y)}\}\}\ \text{m}.$$

Only $N_c \geq 2$ CIDs are scored. Region A has 92,232 raw records and 911 normalized raw-CID values; ten singleton values are excluded, leaving 92,222 records supporting 901 scored raw-CID values.

Direction uses geometric displacement in the original acquisition order of the already coordinate-de-duplicated records. For consecutive retained projected sampling positions $\boldsymbol{p}_j, \boldsymbol{p}_{j+1}$ within the same trace, retain a displacement of at least 1 m and form

$$\boldsymbol{u}_j = \frac{\boldsymbol{p}_{j+1} - \boldsymbol{p}_j}{\|\boldsymbol{p}_{j+1} - \boldsymbol{p}_j\|_2}.$$

Assign the vector to a CID observed at both positions. No vector crosses a trace boundary. No timestamp, elapsed-time threshold, or speed filter is used. For CID c, let $\mathcal{U}_c$ be its assigned unit displacements. When $\mathcal{U}_c$ is nonempty, define $\bar{\boldsymbol{u}}_c = |\mathcal{U}_c|^{-1}\sum_{\boldsymbol{u}\in\mathcal{U}_c}\boldsymbol{u}$, and $R_c = \|\bar{\boldsymbol{u}}_c\|_2$. Direction is reliable exactly when $|\mathcal{U}_c| \geq 3$ and $R_c \geq 0.50$; in that case $\mathrm{dir}(c) = \bar{\boldsymbol{u}}_c/R_c$. Otherwise the directional component is omitted as specified in Table S1.

Canonical source-directed edges are sampled at 5-metre arc-length intervals, including both endpoints. With sample set $Q(e)$,

$$S_d(e,c) = \exp[-d(e,\Omega_c)^2/(2\sigma_c^2)],$$
$$S_h(e,c) = |\{q \in Q(e) : \Omega_c \text{ covers } q\}|/|Q(e)|,$$
$$S_o(e,c) = \{1 + \mathrm{dir}(e)^{\mathsf{T}}\,\mathrm{dir}(c)\}/2.$$

Table S1 completes the operational contract. An unlisted edge–CID pair has score zero.

Graph and endpoint embedding.: For each ordered node pair, retain the shortest directed edge and break an exact tie by the lexicographically smallest original edge identifier. Order outgoing edges by original edge identifier, target, and canonical identifier. Actions occupy 13 fixed slots; absent and hard-infeasible slots are masked, and graphs with larger canonical out-degree are rejected. Start is embedded before goal by deterministic projection onto the successively updated directed graph. Projection ties choose the earliest geometry segment and then the smallest canonical edge identifier. A reverse edge is also split only when its independent projection lies within 5 m. Virtual connector segments add no off-road length and inherit their own source-directed edge identity.

A.2 Evaluation Protocol and Edge Cases

Evaluation removes virtual connectors and restores canonical directed source edges. Road-edge precision/recall use shared predicted/reference edge identities; length-weighted variants replace counts with the corresponding edge lengths. F1 is the harmonic mean. An empty or missing recovery contributes zero to all overlap metrics, whereas a nonempty predicted edge sequence that fails complete O–D validity ($S_{\mathrm{OD}} = 0$) is still scored for road-edge and length-weighted overlap against the reference path. $S_{\mathrm{OD}} = 1$ only for a nonempty, directed-contiguous path connecting the embedded endpoints. All metrics are computed per trajectory from unrounded identities and lengths and then macro-averaged over all 250 held-out queries.

Paired confidence intervals resample the same 250 query indices with replacement for both methods 10,000 times and take the 2.5 and 97.5 percentiles of the macro-score difference.

B. Exact 20-Component SVAA-Aware PPO State

After every appended edge, recompute the SVAA witness. The next pointer is the first observation after the final match, or a completion sentinel. The local observation window spans two positions before through eight after that pointer. Let the distance scale be the larger of 1 m and initial straight-line start–goal distance. Table S2 defines the raw state; every component is clipped to $[-1, 1]$, normalized with the saved training mean and variance plus 10^{-8}, and clipped to $[-10, 10]$.

C. Reward Aggregation and Training Boundary Audit

For a valid step, sum the terms in the main paper's training reward and episode-boundary table (Section III-D) and clip to $[-8, 12]$. Lookahead and bounded-search terms have total-reward weights 0.8 and 0.5; other terms have unit weight. An invalid, masked, or dead-end action instead returns exactly -1 and terminates.

The component rules, path regularization, masks, termination, and truncation conditions are specified once in main-paper Section III-D. The aggregation weights and clipping above complete that contract. Training termination must not be substituted for the exact online endpoint acceptance test in Section F.

D. PPO Optimization and Artifact Binding

Stop the counter exactly at 5,000,000 interactions. Count a final incomplete collection but do not update from it. Region A and B share one fixed evidence, graph/action, state, reward, PPO, search, temporal, corridor, and ranking configuration. They differ only in regional records, graphs, generated-training and held-out-test sets, fingerprints, and learned PPO/normalization values. For each region, model fitting uses only its 1,000 generated training trajectories. Development-time hyperparameter selection uses only the

TABLE S1
OPERATIONAL ROAD–CELL-ID EVIDENCE AND ARTIFACT CONTRACT.

Item	Deterministic rule
Normal evidence	$E_{\mathrm{CID}} = (1.0S_d + 0.8S_h + 1.8S_o)/3.6$. For an unreliable CID direction, set its direction weight to zero and use $(1.0S_d + 0.8S_h)/1.8$.
Compatibility	(e, c) is compatible iff cached $E_{\mathrm{CID}}(e, c) > 0.18$; the strict comparison uses the unrounded value.
Sparsification	For every $N_c \geq 2$ CID, rank all frozen source edges by decreasing unrounded score and then canonical edge identifier; keep $\min(300, \lvert E\rvert)$.
Geometry	Use the frozen, source-oriented geometry and exact 5-metre samples. Boundary samples count as support hits. Reject zero-length projected geometry.
Artifact binding	Require schemas `paper-ecid-v1`/`paper-ecid-raw-v1`, generator constants, projection origin, per-CID parameters, sorted score map, normalized-record hash, graph and source-edge hashes, generator hash, and software versions. Query split edges inherit only their source edge's score map.
Validation	Reject missing or nonfinite metadata, noncanonical/conflicting edge keys, more than 300 source edges for a CID, graph/fingerprint mismatch, or any score outside $[0, 1]$.

TABLE S2
EXACT RAW 20-COMPONENT PPO STATE BEFORE CLIPPING AND NORMALIZATION.

No.	Feature	Construction
1	Goal distance	Divide by the distance scale, cap at one, map to $[-1, 1]$.
2–3	Goal bearing	Normalized longitude and latitude components of the current-to-goal vector.
4	Direction similarity	Cosine between latest movement and goal direction; zero initially.
5–6	SVAA progress	Witness cardinality divided by observation count, and latest cardinality increment divided by that count.
7	Search depth	Current edge count divided by observation count.
8	Last-edge match	Fraction of the latest edge's compatible labels also present as distinct values in the local window; zero initially.
9	Out-degree	Canonical out-degree divided by the frozen graph's actual maximum canonical out-degree.
10–12	Path status	Step/1500; path length/distance scale; and repeated nodes among the latest 12 nodes divided by 12.
13	One-step potential	Maximum distinct observed compatible-label count over canonical successors, multiplied by 0.2 and capped at one.
14	Three-step coverage	Breadth-first search to depth three, first eight successors per node and at most 80 nodes; divide distinct observed labels found by distinct labels in the query.
15	Local target density	Over the first ten successors, sum per-edge counts of compatible target-window labels; divide by observation positions and cap at one.
16	Current-edge evidence	Half the largest evidence between current edge and a distinct local-window label; zero initially or for an empty window.
17–18	Bounded search	Eight-simulation, depth-six value; and one minus the capped standard deviation of its latest eight values.
19	Stagnation	Consecutive nonprogress steps divided by 50.
20	Unique visits	Distinct visited nodes divided by visited nodes.

TABLE S3
FIXED PPO CONFIGURATION SHARED BY BOTH REGIONS.

Item	Value
Model	`MaskablePPO`; separate Actor and Critic branches of widths 256 and 128 with Tanh; normalized 20-vector input; 13 action slots.
Budget	Exactly 5,000,000 interactions in four environments; 1024 steps per environment and 4096 transitions per full collection; batch size 512.
Update	Discount 0.99; GAE 0.95; PPO clip 0.20; learning rate 3×10^{-4}; five epochs; entropy coefficient 0.03; value coefficient 0.5; maximum gradient norm 1.0; target KL 0.03.
Normalization	Normalize observations, not rewards; observation clip 10; stabilizer 10^{-8}.
Binding	Checkpoint, normalization mean/variance, graph, evidence cache, training manifest, and configuration manifest form one fingerprint-matched inference artifact.

Region A generated corpus; none of the 250 real test trajectories is used for model selection. Region B reuses this Region-A-selected configuration without region-specific retuning.

E. TRAINING INPUTS AND ORACLE ARTIFACT AUDIT

Calibration-derived policy-training trajectories.: All reported training trajectories are generated offline from the frozen regional calibration corpus and road graph, without additional training drives. For each region, we randomly select 1,000 O–D coordinate pairs from the calibration data, generate a route and longitude–latitude sequence on the graph, and match every generated coordinate against the calibration records to assign its normalized raw-CID token. The resulting GPS^*–CID sequences supervise PPO; GPS^* denotes generated route coordinates, not measured GPS from a training drive. Sufficient real coordinate–CID trajectories could use the same interface, but all reported policies use the generated sets. For Region A, these 1,000 trajectories are also the sole development set used to select model and method hyperparameters. The 250 field-collected test trajectories remain excluded until the configuration is fixed.

Oracle and lookahead implementation boundary.: The full deterministic oracle construction and bounded lookahead algorithm are given in main-paper Section III-D. Reproduction must bind their graph iteration order, node-pair cache, canonical action slots, and generated training inputs to the artifacts in Section D. The offline oracle contributes reward sets only; query decoding cannot access its coordinates. The following flow records how these components enter the optimization lifecycle.

Offline training flow. Validate graph actions and evidence; construct the training-only generated-coordinate oracle; run four parallel environments; after every accepted edge recompute the SVAA witness, raw state, and bounded-search value; apply masking, reward, and episode boundaries; normalize observations; update MaskablePPO to the exact interaction budget; save the model, normalizer, graph, evidence, and manifests as one bound artifact.

Fig. S1. Complete PPO policy-training procedure.

F. Online Time-Aware SVAA/PPO-Guided MCTS

For a nonempty query sequence, define the trip duration

$$\Delta t_{\mathrm{trip}} = \begin{cases} \max\{t_n - t_1, 1\ \mathrm{s}\}, & n \geq 2,\ t_1, t_n \text{ finite}, \\ 1\ \mathrm{s}, & \text{otherwise}. \end{cases}$$

The decoder rejects an empty observation sequence. The same Δt_{trip} is used by the partial-path cap and the complete-trip filter below; timestamp ratios used by the soft corridor have their own fallback rule.

F.1 Legal Expansion, Prior, Rollout, and Backup

An action is legal only if its target can reach the goal, its projected completion satisfies the temporal length cap in F.2, its prefix remains within three times the embedded shortest-road distance, and its target has appeared fewer than three times. Use maximum depth 500 and stop after 50 steps without either a new match or 5 m best-goal improvement.

For legal action a, the unnormalized action logit is

$$\begin{aligned} z(a) &= 0.55A_a + 0.05E_a - 4R_a \\ &\quad + \frac{0.25}{0.90}\big(0.10G_a + 0.18C_a \\ &\qquad + 0.20T_a + 0.24H_a\big), \\ P(a \mid s) &= \frac{\exp z(a)}{\sum_{b \in \mathcal{A}(s)} \exp z(b)}. \end{aligned} \tag{1}$$

Here A_a is the frozen masked-Actor probability. G_a is the decrease in shortest remaining road distance divided by the previous distance (denominator at least 1 m), clipped to $[0, 1]$. E_a is the larger cached score of the action edge for the observation at the current pointer and, when present, its immediate successor; at the last observation only that observation is used, and at the completion sentinel $E_a = 0$. C_a is the action-level corridor support defined in F.2. T_a is the new pair-time support when the action increases witness cardinality and thereby creates an adjacent matched pair; an interval extension gives $T_a = 0$. H_a is the mean successor lookahead value in state components 13–15, and R_a indicates that the target node already appears in the path history. These definitions reproduce the implemented coefficient construction without an additional prior normalization.

At a rollout endpoint, let $L_0 = \max\{L_{\mathrm{shortest}}, 1\ \mathrm{m}\}$, let d_{rem} be its shortest remaining road distance, and let L_{prefix} be its accumulated length. The previously implicit leaf terms are

$$\begin{aligned} S_{\mathrm{goal}} &= \mathrm{clip}\left(1 - \frac{d_{\mathrm{rem}}}{L_0}, 0, 1\right), \\ S^{\mathrm{leaf}}_{\mathrm{detour}} &= \mathrm{clip}\left(\frac{L_{\mathrm{prefix}}/L_0 - 1}{2}, 0, 1\right). \end{aligned} \tag{2}$$

The interval witness, edge evidence, and raw corridor utility are recomputed at that endpoint before applying the leaf formula in Table S4; V_ϕ is clipped to $[-1, 1]$.

F.2 SVAA, Temporal Constraints, and Corridor

Canonical SVAA witness.: For path $P = (e_1, \ldots, e_m)$ and nonempty observation sequence $C = (c_1, \ldots, c_n)$, initialize $D_{0,i} = D_{q,0} = 0$ and compute

$$\begin{aligned} D_{q,i} = \max\big\{ & D_{q-1,i},\ D_{q,i-1}, \\ & D_{q-1,i-1} + \mathbf{1}[c_i \in \Phi(e_q)] \big\}. \end{aligned} \tag{3}$$

Thus $K = D_{m,n}$. Backtracking follows only predecessors attaining the cell optimum and uses the fixed priority compatible diagonal, edge skip, then observation skip. Reversing the backtracked list yields ordered anchors $((i_r, a_r))_{r=1}^{K}$.

For anchor a_r, the permitted expansion bounds are the first edge when $r = 1$, otherwise $a_{r-1} + 1$, and the final edge when $r = K$, otherwise $a_{r+1} - 1$. Within those bounds, take the maximal contiguous compatible run containing a_r. Process anchors left-to-right; if raw runs overlap, move the later run's start to one edge after the previously assigned end. The earlier observation therefore owns the overlap, every anchor remains inside a nonempty interval, and two observations never reuse an edge position. Unmatched observations and connecting edges remain allowed. Expanding an interval changes its boundaries but never K. This same canonical extractor is used by state construction, search, and ranking; it does not choose a different same-cardinality witness to improve evidence or time for an individual candidate. Interval evidence is zero for $K = 0$; otherwise it averages edges within each interval and then intervals equally. Sequence coverage is $S_{\mathrm{seq}} = K/n$.

Temporal scopes.: For adjacent intervals $(i_r, [\ell_r, u_r])$ and $(i_{r+1}, [\ell_{r+1}, u_{r+1}])$, include both boundary edges and define the span speed in kilometres per hour as

$$\widehat{v}_r = 3.6\, \frac{\sum_{h=u_r}^{\ell_{r+1}} l(e_h)}{\max\{t_{i_{r+1}} - t_{i_r}, 1\ \mathrm{s}\}}.$$

Soft pair compatibility. When either timestamp is nonfinite or the gap is nonpositive, set $\kappa_r = 0$. Otherwise,

$$\kappa_r = \begin{cases} \exp[-(\widehat{v}_r - 50)^2/(2 \cdot 10^2)], & 40 \leq \widehat{v}_r \leq 60, \\ 0, & \text{otherwise}. \end{cases}$$

$$\bar{S}_{\mathrm{pair}} = \begin{cases} 0, & K < 2, \\ \frac{1}{K-1} \sum_{r=1}^{K-1} \kappa_r, & K \geq 2. \end{cases}$$

The exact alignment score is

$$S_{\mathrm{align}} = 0.70\, S_{\mathrm{seq}} + 0.30\, \bar{S}_{\mathrm{pair}}.$$

TABLE S4
ONLINE SEARCH CONFIGURATION.

Item	Deterministic rule
Action prior	Compute the exact legal-action logit in Eq. (1) and softmax only over legal actions. The Critic has coefficient zero in this prior and is used only in leaf evaluation.
Root/PUCT	Mix 92% model prior with 8% Dirichlet noise of concentration 0.25 when multiple root actions exist. Elsewhere use no noise. PUCT constant is 1.55; exact ties use the smallest canonical edge identifier.
Expansion	Instantiate every legal child (at most 13), ordered by decreasing prior then canonical ID. Sample uniformly with probability 0.08; otherwise raise priors to 0.625, renormalize, and sample.
Rollout/backup	Greedily roll out at most 6 off-tree steps by online prior. Evaluate the rollout endpoint and back up one undiscounted scalar running mean. Rollout edges never enter the output tree.
Leaf value	$18S_{\text{align}} + 14S_{\text{edge}} + 8.5S^{\text{raw}}_{\text{corridor}} + 2S_{\text{goal}} - 6S_{\text{detour}} + 0.15\ \text{clip}(V_\phi, -1, 1)$.
Budget	30,000 simulations per query.

This interval-span kernel is a soft protocol proxy, not an instantaneous speed estimate or a hard feasibility rule. Thus the 40–60 km/h gate changes only the pair score and never prunes a candidate.

Partial-path hard pruning. Prefix length plus shortest remaining road distance not to exceed Δt_{trip} times 22.222222 m/s (80 km/h). *Complete-trip hard feasibility.* Complete candidates must lie within Δt_{trip} times 0.975 to 24.3 m/s, the configured 1.5–18.0 m/s range with 35% tolerance. The stricter generator cap therefore determines the effective upper bound.

Corridor implementation boundary.: Main-paper Section III-D specifies the deterministic window aggregation, reliability thresholds, path placement, action-level support, and raw corridor utility. The same frozen construction is used here without a second parameterization. Corridor support changes priors and scores; it never hard-prunes, repairs, or completes a path. The temporal feasibility tests above remain separate from this soft evidence, and final normalization is confined to each method's own retained candidate set as specified below.

F.3 Complete Candidates, Normalization, and Final Ranking

Traverse MCTS children in canonical depth-first order after 30,000 simulations. Keep only complete tree-generated goal paths satisfying the trip filter and deduplicate by full canonical edge sequence. Do not complete prefixes with shortest paths, corridor edges, or rollout edges. If none remain, report failure. Normalize only raw corridor utility within the query's retained candidate set: use min–max normalization when its range exceeds 10^{-12}, and otherwise clip each raw value to $[0, 1]$. With $L_0 = \max\{L_{\text{shortest}}, 1\ \text{m}\}$, define $S_{\text{detour}} = \text{clip}((L/L_0 - 1)/2, 0, 1)$,

$$\begin{aligned}\text{Score}_{\text{final}} = S_{\text{align}} &+ 0.75S_{\text{edge}} \\ &+ 0.75S^{\text{norm}}_{\text{corridor}} - 0.45S_{\text{detour}}.\end{aligned}$$

The final score contains neither Critic, an extra speed term, nor an evaluation metric. Exact ties choose the lexicographically smallest complete canonical edge sequence.

G. CONTROLLED COMPARISONS AND COMPACT RESULTS

G.1 Baseline and Ablation Definitions

Baselines.: O–D Dijkstra uses endpoints only. Yen K-Shortest Paths generates at most 200 loopless paths; Beam Search retains width 128 and uses a 30,000-state expansion cap. Both use the common road–CID evidence, SVAA witness, temporal constraints, and final route score specified below, without learned Actor priors or Critic values. CID-ST-HMM uses support emissions and timestamp-, reachability-, and endpoint-aware Viterbi. CID-MMA Transformer predicts candidate-road probabilities followed by graph-constrained Viterbi. The three ODT adaptations in Table S5 retain selected architectural components but change information sources, supervision, or decoding. They are not native-system reproductions.

Adaptation scope.: The following implementations retain selected architectural components but replace native information sources or supervision to fit the ODT task. They are not reproductions of the original published systems.

Common route-evaluation contract.: Yen, Beam, and the proposed decoder share the frozen E_{CID}, strict compatibility threshold 0.18, canonical SVAA witness in F.2, pair-time kernel, CID–time corridor, and detour definition. Let $\mathcal{C}_A(X)$ be the retained feasible complete-candidate set of method $A \in \{\text{Yen}, \text{Beam}, \text{MCTS}\}$. All three methods rank their complete candidates using the F.3 score:

$$\begin{aligned}\text{Score}_{\text{final}}(P; \mathcal{C}_A) = 0.70S_{\text{seq}} + 0.30\bar{S}_{\text{pair}} + 0.75S_{\text{edge}} \\ + 0.75S^{\text{norm}}_{\text{corridor}}(P; \mathcal{C}_A) - 0.45S_{\text{detour}}.\end{aligned}$$

Raw corridor utility is normalized within each method's own retained complete-candidate set using the F.3 rule. Thus the evidence, alignment, temporal terms, and final coefficients are shared, while each method constructs its own candidates. Yen and Beam use no learned Actor priors or Critic values.

Method-specific budget semantics.: The 30,000 MCTS simulations count selection, expansion/rollout, and backup iterations, not 30,000 complete routes. Yen's $K = 200$ caps generated complete O–D paths, whereas Beam width 128 caps retained partial paths at each depth. These controls are neither candidate-count nor compute equivalents. We therefore compare end-to-end recovery accuracy under pre-frozen configurations tailored for retrospective trajectory reconstruction, reporting query-time latency alongside accuracy to characterize the computational trade-off across operating budgets.

Controlled interventions and ablations.: The first four variants retain the complete method's frozen PPO checkpoint and normalizer and change one decoder-side rule; they measure frozen-guidance sensitivity, not the effect of retraining an alternative policy. *Synchronous Alignment* replaces only SVAA

F.4 End-to-End Textual Pseudocode

Input: directed graph and endpoint-embedding rules; origin and destination; nonempty timestamped raw-CID sequence; frozen evidence cache, PPO Actor/Critic, normalizer, and inference configuration. **Output:** one canonical connected O–D path or explicit failure.

1) Validate all artifact fingerprints and the 13-slot action contract. Embed the start and then the goal, compute L_0 and Δt_{trip}, and build the deterministic CID–time corridor.
2) Initialize the root with an empty edge sequence, zero visits, and its canonical SVAA witness, state, legal-action mask, length, visit history, and stagnation variables; initialize every subsequently created tree edge with $N(s,a) = Q(s,a) = 0$.
3) For each of 30,000 simulations, descend through expanded nonterminal nodes by the main-paper PUCT rule. Terminate a branch at the depth or stagnation limit; otherwise mask actions that violate reachability, projected-completion length, detour, or revisit rules.
4) At the first expandable node, construct the normalized PPO state, evaluate Eq. (1), instantiate every legal child in prior order, mix Dirichlet noise only at the root, and sample one child by the fixed uniform/power-mixture rule.
5) Append the sampled tree edge; update length and visit history; recompute the complete canonical SVAA witness, distinguishing extension of an existing interval from a new matched observation; then update pair-time and stagnation state.
6) Unless terminal, run at most six off-tree greedy steps using the same legal mask and online prior. Rollout edges update the hypothetical state but are never inserted into the output tree.
7) At the rollout endpoint, recompute S_{align}, S_{edge}, $S^{\mathrm{raw}}_{\mathrm{corridor}}$, S_{goal}, $S^{\mathrm{leaf}}_{\mathrm{detour}}$, and the clipped Critic term. Apply the leaf formula in Table S4 and update the running mean V at every visited non-root tree edge by $N \leftarrow N + 1$ and $Q \leftarrow Q + (V - Q)/N$.
8) After the budget, traverse tree children in canonical depth-first order. Retain only tree-generated goal paths satisfying the complete-trip filter and deduplicate by their full canonical edge sequence; report failure if the set is empty.
9) Recompute every retained path's canonical witness and final components. Normalize only $S^{\mathrm{raw}}_{\mathrm{corridor}}$ within this candidate set and apply the F.3 score.
10) Return the highest-scoring path; an exact score tie selects the lexicographically smallest complete canonical edge sequence. No prefix completion, rollout edge, corridor edge, or reference path can enter the returned sequence, and the final ranking contains no Critic term.

Fig. S2. Complete online constrained-decoding procedure.

TABLE S5
ARCHITECTURAL SCOPE OF THE IMPLEMENTED ODT ADAPTATIONS.

Adaptation	Retained components	Adapted information or decoding
DMM-ODT	CID/road embeddings, GRU encoder–decoder, global attention	Regional road supervision, O–D conditioning, timestamp-aware decoding, and graph stitching replace parts of the native pipeline.
CID-LHMM-ODT	Multi-relational representations, learned probabilities, Viterbi decoding	Calibration-derived observations replace tower-position information; O–D/time constraints and graph stitching are added.
ZSMM-ODT-Sup	Two-channel raster representation, U-Net calibration, constrained path solving	Calibration-derived CID positions and regional road supervision replace the native zero-shot evaluation setting.

with position-wise matching: road and observation positions advance together, without gaps or multi-edge intervals. *One-CID-One-Edge* retains, for each CID, only its highest-evidence compatible road edge (canonical-edge tie-break). *Point/Centroid Representation* replaces each empirical support Ω_c by the mean accepted coordinate and rebuilds compatibility. *No Set-Valued Alignment* instead retains, for each edge, only its highest-evidence compatible CID (CID-string tie-break), while a CID may still label many edges. Thus the latter two relation collapses are not duplicates. *Pure PPO* removes MCTS, rollouts, and complete-tree comparison and greedily selects the highest-probability legal Actor action. *Pure MCTS* retains the same tree, budget, hard rules, extraction, and ranking but removes Actor priors and Critic leaf values. *No Timestamp* removes every timestamp-derived online speed, trip, corridor-placement, leaf, and ranking term while retaining CID order. All unlisted graph, split, endpoint, metric, budget, seed, and tie settings remain fixed.

G.2 Region A Protocol and Results

Region A is field-collected in an urban district of a southern Chinese city (City A) using professional drive-test equipment that synchronously logs GPS coordinates and scanned Cell-ID sequences over the city's real directed road network. It contains 92,232 field-collected calibration records covering 911 normalized raw-CID values and a graph with 6,277 nodes and 12,777 directed edges. Its 1,000 policy-training trajectories are generated from those records and the graph under Section E, requiring no additional training drives; they are the only trajectories used for policy fitting and hyperparameter selection. All 250 test trajectories are real vehicle drives recorded with the probe and held out until the model and hyperparameters are fixed. They are used only for final held-out evaluation and never form a model-selection split. A different researcher collected them without access to the sampled training O–D pairs, generated routes, CID sequences, or other training-set information. Test records do not enter calibration, training-data generation, policy training, or hyperparameter selection. Test GPS, CID, and timestamps were recorded synchronously. Evaluation GPS polylines are converted independently of predictions into canonical road-edge references by approximately 5-metre densification, nearest-node mapping, and directed length-weighted shortest paths between nonadjacent mapped nodes. These automatically constructed references are used only after prediction and may retain projection uncertainty at parallel or grade-separated roads.

For a predicted edge sequence E_p and reference sequence E_r, $F_{\mathrm{LCS}} = 2\,\mathrm{LCS}(E_p, E_r)/(|E_p| + |E_r|)$; unlike edge-set F, it preserves traversal order and repeated nonconsecutive edges.

TABLE S6
MEAN REGION A RESULTS OVER ALL 250 HELD-OUT TRAJECTORIES (%). F_{LCS} IS THE ORDER-SENSITIVE ROAD-EDGE SEQUENCE LCS-F1.

Method	P	R	F	P^L	R^L	F^L	S_{OD}	F_{LCS}
Ours	**90.25**	**95.30**	**92.47**	**92.08**	**96.03**	**93.83**	**100.00**	**88.95**
O–D Dijkstra	84.94	61.97	69.85	84.87	61.18	68.81	99.60	75.37
Yen KSP + score	75.68	61.82	66.31	76.39	60.77	65.60	99.60	67.27
Beam Search + score	80.31	67.85	71.79	80.11	66.89	70.79	99.60	74.34
CID-ST-HMM	27.23	19.71	18.92	23.92	20.59	18.24	46.40	29.56
CID-MMA Transformer	26.26	20.29	18.65	24.68	21.00	18.70	45.60	30.18
DMM-ODT	55.00	32.40	37.32	53.38	30.51	34.80	100.00	56.20
CID-LHMM-ODT	63.87	41.33	45.54	60.00	42.96	45.57	100.00	67.41
ZSMM-ODT-Sup	70.17	61.21	61.74	74.54	55.77	59.93	100.00	51.84
Synchronous Alignment	74.44	71.59	71.52	74.90	70.90	70.82	100.00	69.40
One-CID-One-Edge	77.27	70.98	72.19	77.75	71.04	72.10	100.00	69.34
Point/Centroid	81.23	76.37	76.72	82.12	76.10	76.68	100.00	75.15
No Set-Valued Alignment	77.24	69.86	71.65	78.26	70.05	71.93	100.00	71.11
Pure PPO	31.93	64.60	38.35	31.42	63.39	37.26	87.20	15.76
Pure MCTS	76.72	77.32	75.61	77.43	77.13	75.59	89.20	15.93
No Timestamp	78.59	56.74	58.88	77.66	57.07	58.66	90.40	15.42
No-GPS Training	82.38	94.26	86.65	83.58	94.68	87.31	100.00	81.29

The discrepancy between road-edge F1 and FLCS reflects their different evaluation objectives. Road-edge F1 measures edge-set overlap regardless of traversal order, whereas FLCS additionally penalizes ordering discrepancies and repeated traversals. Pure MCTS achieves relatively high edge coverage but substantially lower sequence agreement (75.61% road-edge F1 versus 15.93% FLCS), indicating that its recovered paths may contain many relevant road edges without preserving their correct traversal order. This observation further highlights the importance of learned guidance for complete-path reconstruction.

The additional *No-GPS Training* control removes the GPS^*-derived oracle reward during policy training while retaining the calibrated road–CID evidence and endpoint inputs. A separate PPO checkpoint and observation normalizer are fitted for this training setting and frozen before evaluation. The online query inputs, constrained decoder, and evaluation protocol remain unchanged.

G.3 Search-Budget Sensitivity and Computational Cost

We evaluate PPO-guided MCTS and Pure MCTS under six simulation budgets from 1,000 to 30,000 on the same 250 Region A test trajectories. Table S7 reports the complete performance metrics, mean inference latencies, and accuracy advantages across all budgets.

TABLE S7
SEARCH-BUDGET SENSITIVITY AND INFERENCE LATENCY COMPARISON ON 250 REGION A TEST TRAJECTORIES.

Budget	Guided F1	Guided Time	Pure F1	Pure Time	Gain (pp)
1,000	72.00%	4.7232 s	64.50%	3.3959 s	+7.50
3,000	78.00%	6.7010 s	67.80%	5.1197 s	+10.20
5,000	82.00%	8.0349 s	70.20%	6.3593 s	+11.80
10,000	87.60%	10.8531 s	73.00%	7.6196 s	+14.60
20,000	91.40%	16.2796 s	74.80%	13.2728 s	+16.60
30,000	92.47%	21.7062 s	75.61%	18.6803 s	+16.86

Beam Search (width 128, 30,000 expansion cap) achieves 71.79% road-edge F1 with a mean query latency of 2.5186 seconds. While Beam Search provides fast heuristic decoding, PPO-guided MCTS demonstrates decisive accuracy superiority, achieving a 20.68-point F1 advantage at 30,000 simulations and outperforming Beam Search in accuracy at all budgets of 1,000 and above. Crucially, at 3,000 simulations, Guided MCTS reaches 78.00% F1 in 6.7010 seconds, exceeding Pure MCTS at 30,000 simulations (75.61% F1, 18.6803 seconds) while slashing mean inference latency by 64.1%.

H. FIGURES REFERENCED BY THE MAIN PAPER

Against Beam Search, paired 10,000-resample 95% intervals are $[18.15, 23.29]$ for road-edge F1 and $[20.23, 25.94]$ for length-weighted F1; Ours wins 232/250, ties one, and loses 17. These intervals reflect test-query sampling uncertainty.

H.1 Fixed Qualitative and Per-Trajectory Audits

Reading guide. The training-only oracle is separate from the PPO state and saved inference artifact. Tree edges can form

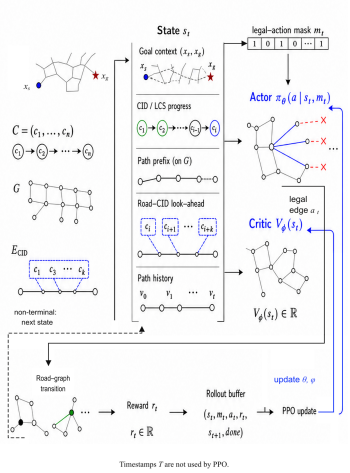


Fig. S3. PPO neighbor-edge policy learning. Timestamps are excluded from state, action, and reward; the Actor, Critic, and normalizer are frozen before online decoding.

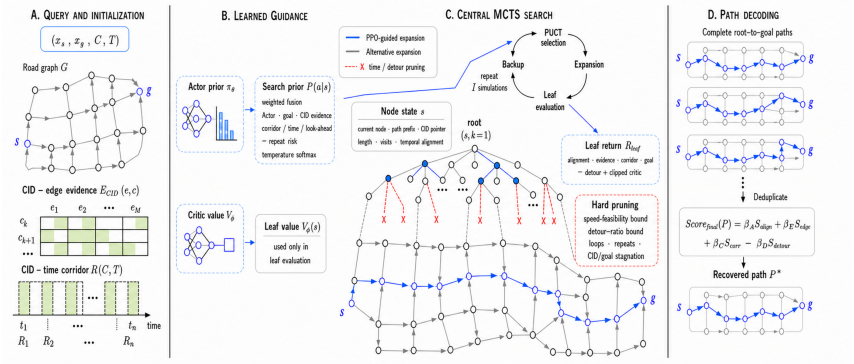


Fig. S4. Online time-aware PPO-guided MCTS. Only complete tree paths are candidates; rollout edges estimate leaf value but never enter an output path, and the Critic does not enter final ranking.

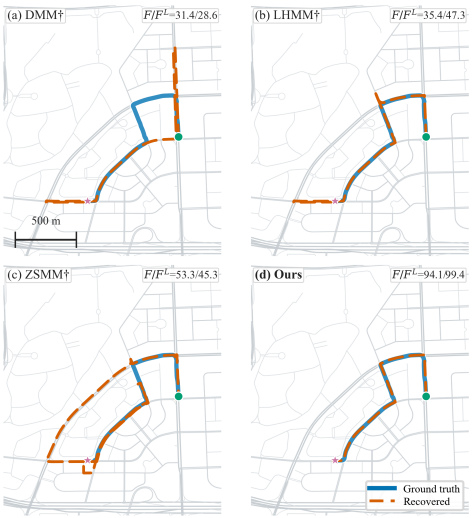


Fig. S5. Same-query comparison on fixed Region A test trajectory 9: DMM-ODT, CID-LHMM-ODT, ZSMM-ODT-Sup, and Ours. Every panel uses the identical query and automatically constructed ground truth; blue is ground truth and red is the method's single returned path. No method is rerun or manually selected for this trajectory.

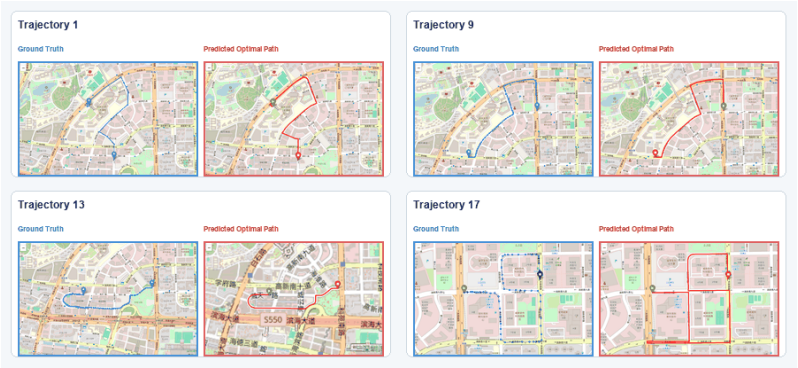


Fig. S6. Reference and recovered routes for fixed Region A examples (trajectories 1, 9, 13, and 17). The prediction is the single returned path from the fixed run; no trajectory-specific rerunning or manual path selection is used.



Fig. S7. Compact per-trajectory audit for Ours and all seven controlled variants on the 250 Region A tests in fixed original order. Columns are identical trajectories across rows; color encodes stored road-edge F1 and a white cross marks O–D failure. The visualization uses all 2,000 per-trajectory records without sorting or selection; the eight-metric macro averages are in Table S6.

candidates; rollout edges only estimate leaf value; corridor edges only provide soft evidence. "Ground Truth" means

the automatically constructed reference; "Predicted Optimal Path" means the single returned candidate from the frozen run. Background maps never enter decoding or metrics.

Controlled interpretation. The four representation/SVAA interventions retain 100% connectivity but lose 15.75–20.95 road-edge-F1 points, separating route fidelity from path feasibility. Pure PPO exposes long-horizon local-policy error; Pure MCTS shows the benefit of learned guidance; and No Timestamp shows that temporal constraints affect recall and feasibility. These controlled changes do not rank native systems under their original supervision.

Connectivity results. O–D Dijkstra, Yen KSP, and Beam Search each return a valid canonical O–D path on 249 of the 250 Region A queries, yielding $S_{\mathrm{OD}} = 99.60\%$. The proposed decoder returns valid connected paths on all 250 queries. All queries, including failed recoveries, contribute to the reported macro-averaged metrics.

I. Cross-City Pipeline Transfer and Controlled Scaling Lab (Region B)

Region A and Region B fulfill complementary roles in the evaluation design. Whereas Region A serves as the primary real-world physical benchmark (four-PLMN drive tests over City A's real road graph), Region B isolates cross-city topological transfer, $3.6\times$ CID-vocabulary expansion, hidden-tower verification, and million-record scaling laws over the real directed road network (2,152 nodes and 4,225 directed edges) of another southern Chinese city (City B). Using industry-standard cellular simulation rules incorporating base-station coverage footprints, signal-matching rules, and cell-handover logic, the external simulator generates 1,316,400 calibration records (3,291 unique raw-CID values) and 250 held-out test trajectories. All underlying simulator parameters—including base-station coordinates, coverage polygons, and handover thresholds—are strictly withheld from our framework, providing an explicit closed-oracle check where ground-truth tower geometry exists yet remains invisible to decoding: our pipeline observes only calibration (GPS, raw-CID) pairs and held-out (O–D, timestamped raw-CID) queries, builds 1,000 generated training routes under Section E, and ensures that simulated test records never enter calibration, training-data generation, policy training, or hyperparameter selection. Together, Region A establishes competitiveness and multipath resilience under genuine four-PLMN drive-test conditions, while Region B proves zero-retuning cross-topology transferability, blinded-tower isolation, unconfounded 10%–100% calibration scaling across 1.32M records, and 200-run multi-seed robustness. Region B's policy is fitted on its 1,000 generated trajectories using the frozen Region A hyperparameters and decoding settings fixed prior to evaluating the 250 test queries, replicating the calibration–training–decoding pipeline on distinct regional artifacts rather than zero-shot weight reuse.

TABLE S8
Scale of the rule-simulated Region B data and City B road graph.

Asset	Count
Generated training trajectories	1,000
Simulated calibration records	1,316,400
Unique raw-CID values	3,291
Road-graph nodes	2,152
Directed road edges	4,225
Rule-simulated test trajectories	250

TABLE S9
Aggregate results for the two separate regional evaluations (%).

Region	P	R	F	S_{OD}
A	90.25	95.30	92.47	100.00
B	96.19	93.94	94.86	100.00

Region	P^L	R^L	F^L
A	92.08	96.03	93.83
B	96.67	94.43	95.34

Evaluation and interpretation.: Region B uses the same per-trajectory macro metrics, failure conventions, canonical-edge restoration, and evaluation-reference construction as Region A. A failed query remains in every denominator. The Region B row does not enter the Region A baseline gains or bootstrap intervals. Region B transfers the frozen Region A hyperparameters and decoding configuration without retuning onto an independent rule-simulated urban road-network benchmark with newly constructed regional evidence and a separately trained policy. Consistent with Region B's denser calibration coverage (1,316,400 vs. 92,232 records) and uniform handover profiles, Region B yields higher macro precision and F1, mutually reinforcing Region A's real four-PLMN field validation by confirming zero-retuning cross-topology transfer and $3.6\times$ CID scaling.

Scaling and multi-seed repeatability audit scope.: The Region B scaling curves and multi-seed repeatability audit are reported in main-paper Fig. 4 and Section IV-E. Calibration subsets (10%–100%) rebuild Region B regional evidence and retrain the policy to verify monotonic gains from denser calibration coverage, whereas the policy-training comparison (50, 250, 500, and 1,000 generated routes) evaluates all trained policies on the exact same 250 Region B test queries with identical evidence cache, road graph, decoder, and search budget. Each training size uses one trained policy; the minor dip at 500 routes reflects training-composition and optimization sensitivity under the fixed 5M interaction budget rather than a causal harm of extra data. For the stochastic robustness audit in Fig. 4(c), 20 test trajectories are randomly drawn from Region B and decoded 10 times each under 10 distinct inference random seeds (200 total runs) with fixed model parameters, input data, and MCTS search budget. All 20 trajectories return identical predicted road-edge sequences across all 10 random seeds, yielding a sample standard deviation of 0.0000 pp across all

160 valid metric–trajectory combinations spanning the seven core Table S9 metrics plus order-sensitive F_{LCS} defined in Section G.2 and indicating stable decoding across these 20 sampled Region B queries under the evaluated configuration. All aggregate effectiveness comparisons use the full 250-query sets.

Qualitative audit.: The complete fixed-order visual audit contains 13 consecutive sheets: 12 sheets of 20 test trajectories and one sheet for trajectories 241–250. Every card uses the same extent for reference and prediction. All 13 sheets, in original test order, are packaged in the supplementary media archive; Figure S8 shows the first and final sheets, selected by position in that fixed order rather than performance.

Regional invariance audit.: The two regional runs share the evidence-grid and buffering constants, direction-reliability rule, sparsification cap and compatibility threshold, graph canonicalization and 13-slot action contract, state and reward definitions, PPO architecture and optimizer settings, training interaction budget, online simulation budget, PUCT and rollout rules, temporal bounds, corridor construction, final coefficients, and deterministic tie rules. No Region B threshold, coefficient, budget, or speed range was selected from its test results. Region-specific quantities are limited to the permitted inputs and their consequences: calibration records, road graph, generated-training and held-out-test sets, cache/hash values, and the PPO weights and normalization statistics learned from those inputs.

Artifact and reference separation.: The Region B evidence cache is built before evaluation from its own permitted rule-simulated calibration corpus (with underlying simulator tower coordinates and handover rules hidden) and is consumed read-only by training and prediction. Its simulated test trajectory GPS polyline is projected to a reference only for post-prediction metric computation. Neither that polyline nor the resulting canonical edges can change search, corridor construction, candidate extraction, or ranking. This is the same separation enforced in Region A and prevents the additional regional experiment from silently changing the method's information boundary.

J. Reproducibility Checklist and Limitations

J.1 Reviewer-Facing Reproducibility Checklist

TABLE S10
Reviewer-facing phase checklist.

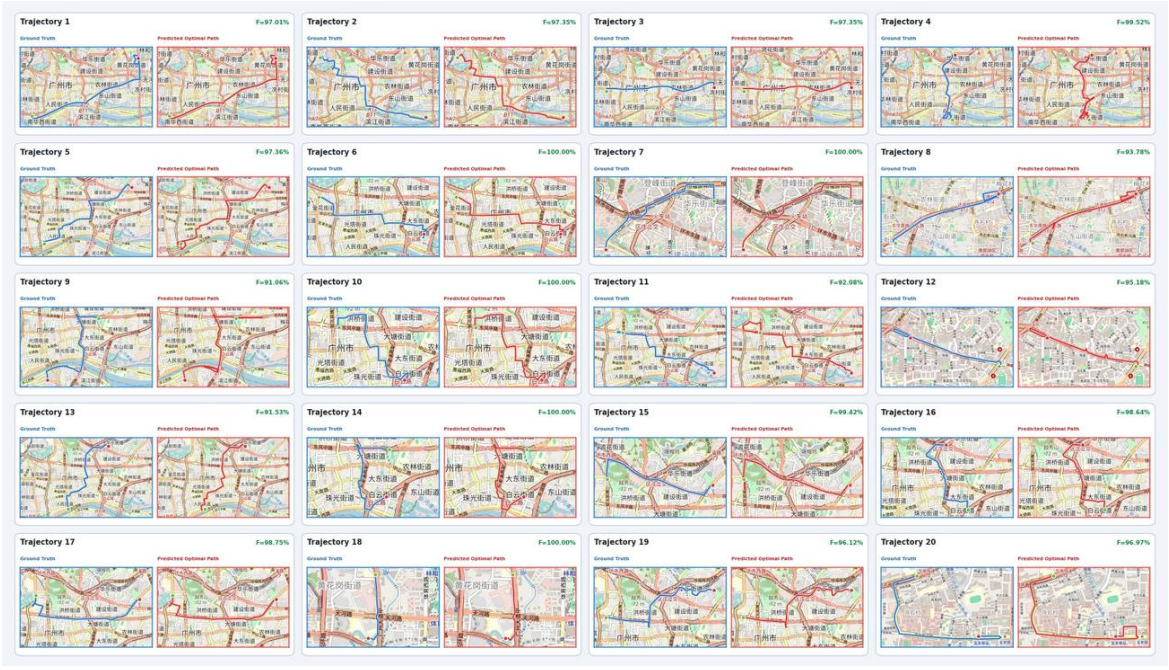
Phase	Fixed boundary
Calibration	Validate the graph, accepted probe GPS–CID rows, generator constants, sorted reductions, and hashes. Exclude tower coordinates/inventory, operator road associations, and evaluation queries.
PPO training	Use the frozen graph/cache, calibration-derived, road-graph-generated training trajectories, training-only oracle, 20-vector state, mask, and exact budget for policy fitting and hyperparameter selection. Exclude all 250 regional held-out tests (physical drives in Region A; rule-simulated routes in Region B) and their timestamps from model selection, state, action, reward, and bounded lookahead.
Online decoding	Use endpoints, timestamped CIDs, graph/cache, checkpoint, normalizer, and fixed search configuration. Exclude intermediate query GPS, evaluation edges, post-hoc completion, and manual repair.
Evaluation	Compare the returned canonical path with an independently constructed held-out reference. Compute per-query metrics before macro-averaging and provide no feedback to any earlier phase. This empirical validation is not a model-selection split.

Every proposed result is bound to a regional evidence fingerprint, graph, PPO checkpoint, matching normalizer, split manifest, and inference configuration. The four representation/alignment interventions reuse the complete method's Region A PPO checkpoint and normalizer and change only the named decoder-side rule. The additional training-supervision control uses its separately fitted checkpoint and normalizer. Displayed percentages are rounded only after aggregation. The separate media archive contains only fixed rendered outputs and a manifest; it does not alter the technical PDF or provide a mutable web pointer.

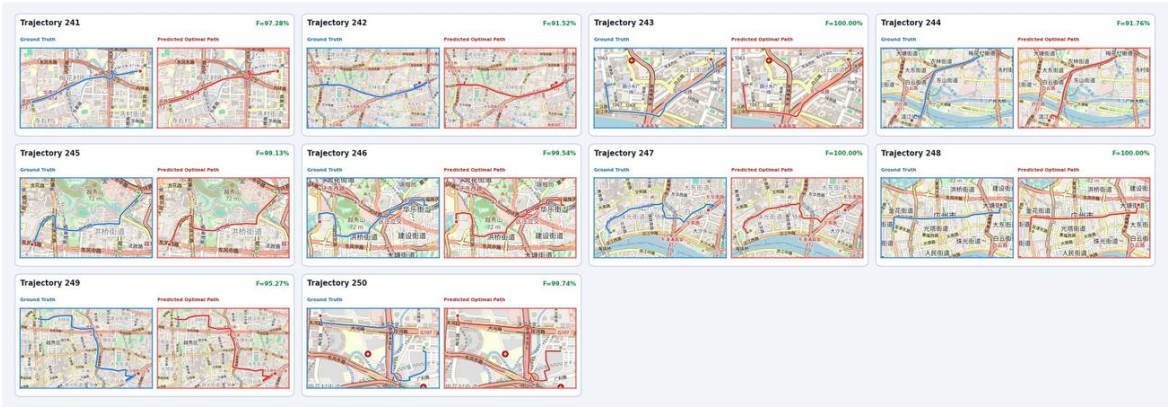
J.2 Limitations and Future Work

The evaluated pipeline uses regional GPS–CID calibration, policy training, known endpoints, and temporal settings fixed for the Region A physical vehicle acquisitions and the rule-simulated Region B benchmark. Future work will also relax explicit O–D endpoint conditioning toward endpoint-agnostic reconstruction (alongside perturbed anchors and multi-city field acquisitions), different traffic and sampling regimes, evidence aging, and reduced-calibration transfer. Metrics use automatically constructed road-edge references; targeted audits can quantify ambiguity near parallel or grade-separated roads. Query-bootstrap intervals and multi-seed inference repetitions characterize distinct sources of statistical and decoding uncertainty, while varying independent training seeds can further extend this assessment.

The Region A field drive-test dataset is available at https://anonymous.4open.science/r/Trajectory_Recovery-223B/; the rule-simulated Region B benchmark corpus (built over City B's road network) and complete implementation code will be publicly released upon acceptance. Released assets will respect applicable map/data licenses and privacy requirements. Region A uses researcher-controlled probe drives rather than passive subscriber exports, whereas Region B is rule-synthesized



(a) Trajectories 1–20



(b) Trajectories 241–250

Fig. S8. First and final fixed-order rule-simulated Region B route sheets (City B). Blue denotes the automatically constructed reference and red the single path from the frozen inference run. The other 11 consecutive sheets are in the supplementary media archive; no query is rerun and no path is manually selected. Pixel overlap is not an evaluation metric; Table S9 is computed from canonical edges and lengths.

without real user trajectories; sensitive timestamps and mobility traces will be handled under the release protocol.